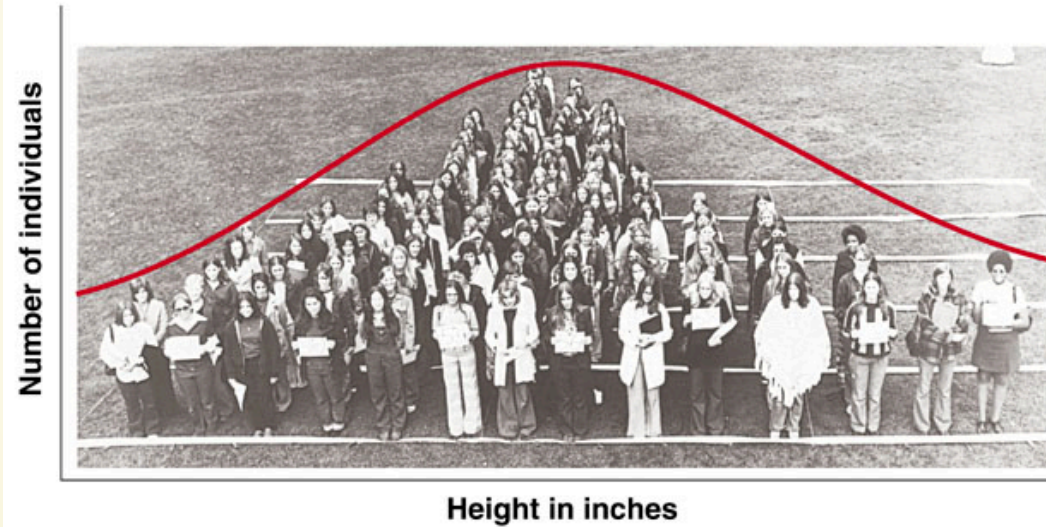
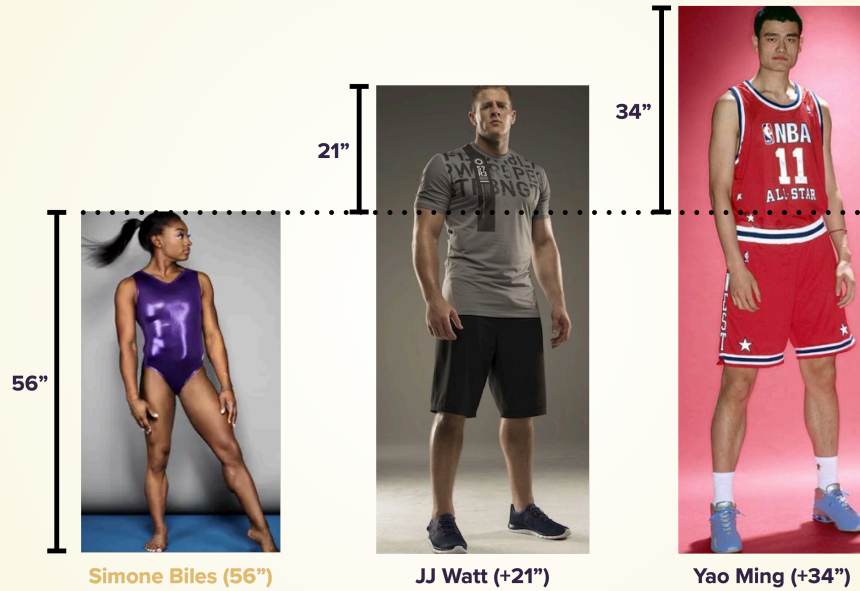


INHERITANCE OF COMPLEX TRAITS

Tobin/Dusheck, Asking About Life, 2/e
Figure 16.6



INHERITANCE OF COMPLEX TRAITS



1. <https://artpictures.club/autumn-2023.html>

QUANTITATIVE GENETICS

? What does it study and what it doesn't?

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- Inheritance of quantitative traits and how they vary in a population

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Is Quantitative Trait Compatible With Mendelian Genetics?

- When Mendelian genetics was rediscovered in the early 1900s, emphasis was placed on one to three loci segregating, creating discrete phenotypic categories. seems incompatible with continuously varying traits (why?)

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1. Introduction to Genetic Analysis, ed 11, p716

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- This made Mendelian genetics initially an obstacle to the acceptance of Darwin's evolutionary theory.
- What do you think?

1. Introduction to Genetic Analysis, ed 11, p716



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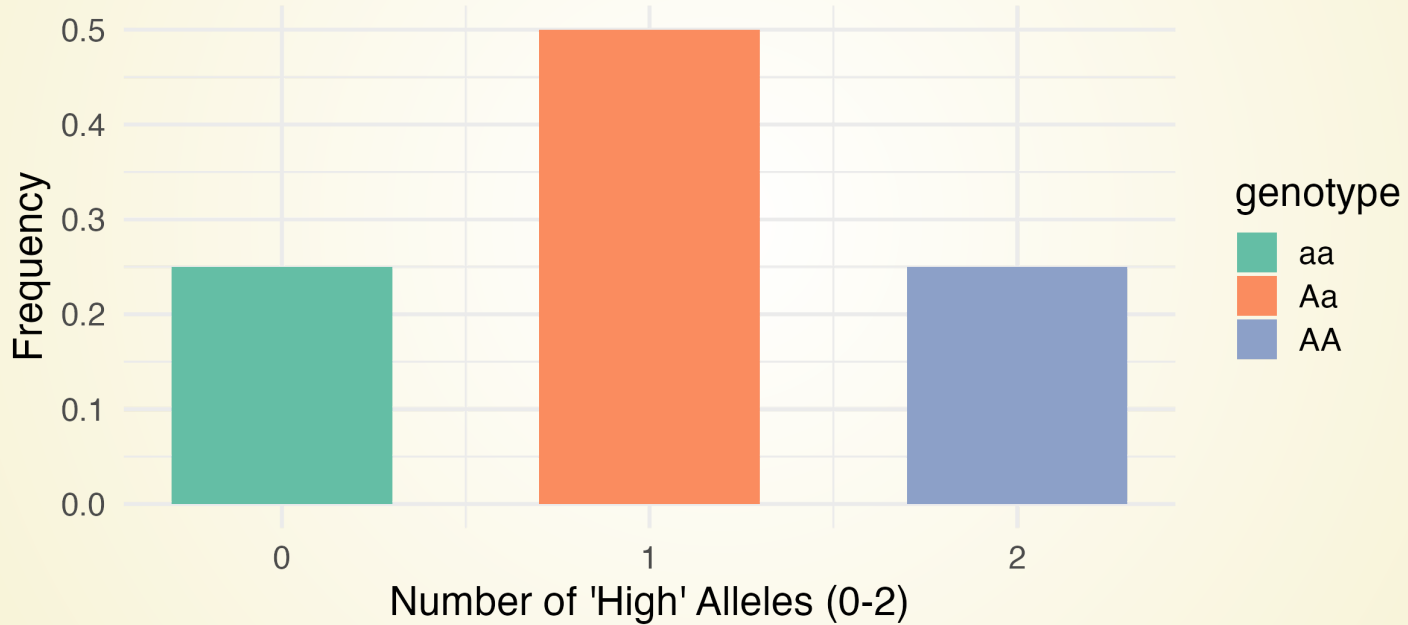
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- How about two loci? Three? Ten?

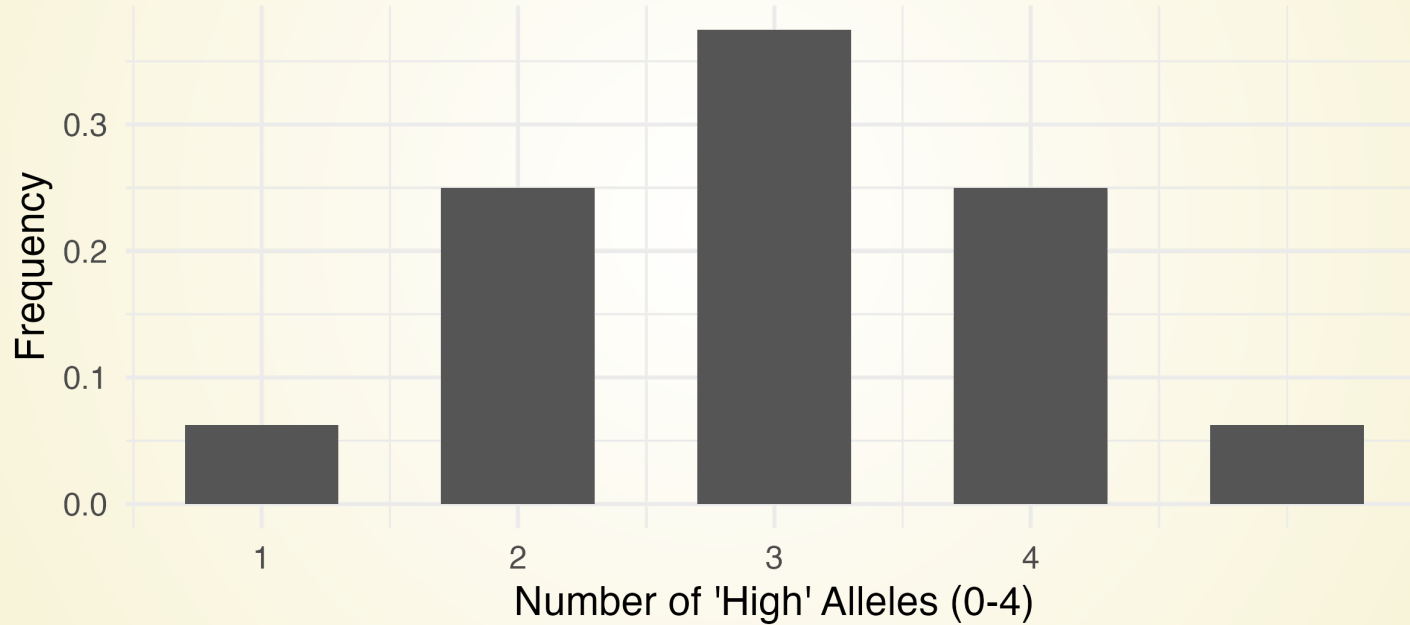
Multifactorial Hypothesis: One Locus

Single Locus (Co-dominance)



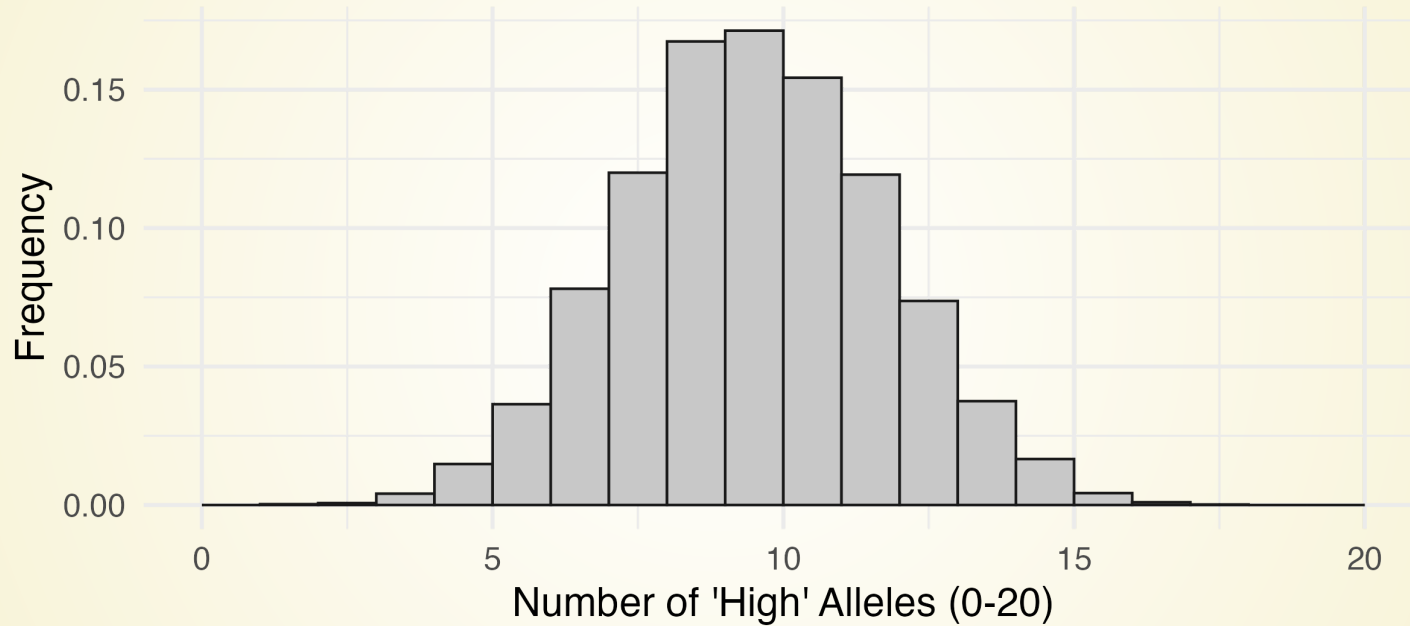
Multifactorial Hypothesis: Two Loci, Additive

Two Additive Loci (Co-dominance)



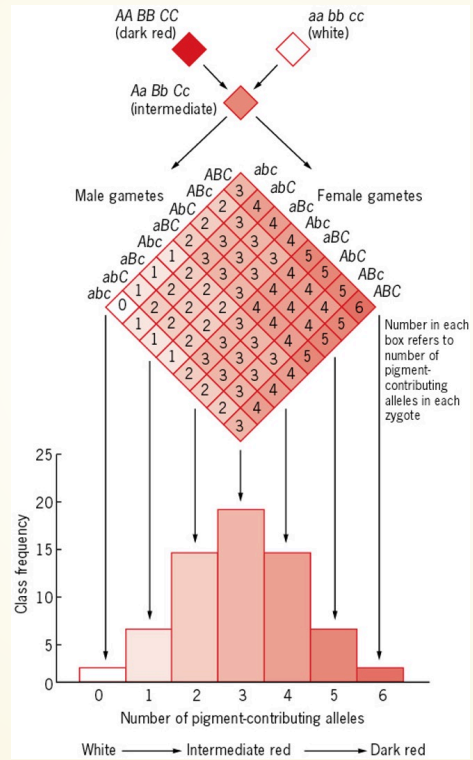
Multifactorial Hypothesis: Ten Loci, Additive

Polygenic Trait (10 Additive Loci, Co-dominant)



1. code [here](#)

Multifactorial Hypothesis



Define The "Genetic Architecture" Of A Quantitative Trait

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Types Of Traits And Inheritance

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 - typically have complex inheritance involving multiple genetic and environmental factors.
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 - typically controlled by one or two genes, hence exhibiting simple inheritance

Types Of Traits And Inheritance

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 - having multiple genetic and environmental factors
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- **Meristic trait**, or "counting trait", can take on a number of discrete values.

Statistical Description Of A Quantitative Trait

Mean: the arithmetic average of a set of data

$$\bar{X} = \frac{\sum X_i}{n}$$

Variance is a measure of the deviation from the mean

$$\sigma_X^2 = E[(X - \mu)^2] = E[X^2] - E[X]^2$$

$$\sigma_X^2 = \frac{1}{n} \sum (X_i - \bar{X})^2$$

$$s_X^2 = \frac{1}{n-1} \sum (X_i - \bar{X})^2$$

More On Mean And Variance

- population mean, sample mean, expectation of random variable
-
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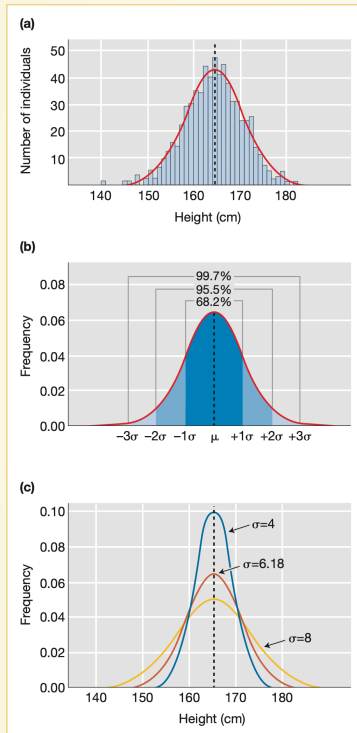
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More On Mean And Variance

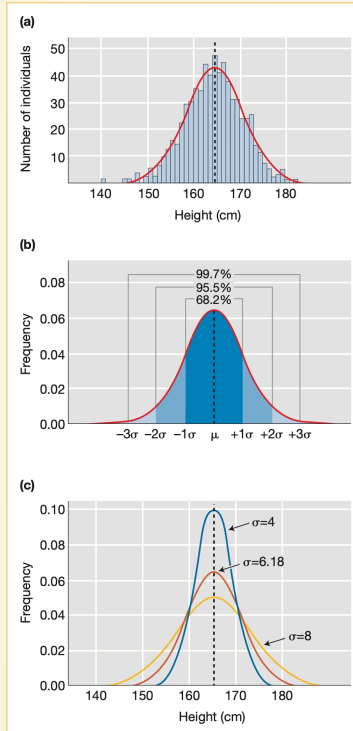
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- What's the relationship and difference between sample variance, sample standard deviation (SD) and standard error of the mean (SEM)?

Normal Distribution

- $X \sim N(\mu, \sigma^2)$

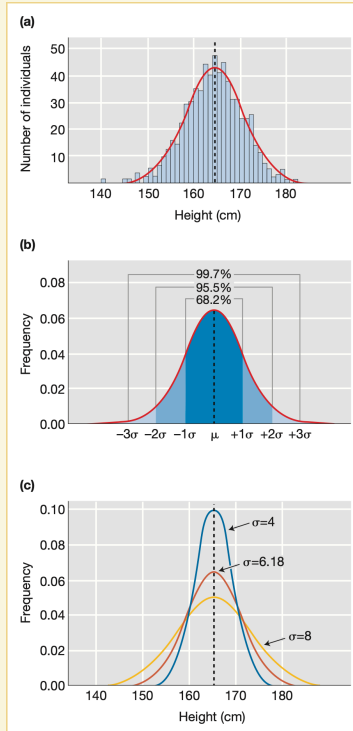


Normal Distribution



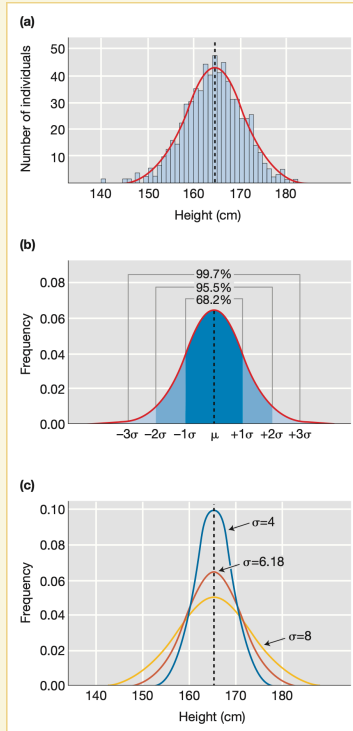
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- Can mean and variance vary independently?
 - is that generally true of all distributions?

A Simple Genetic Model For Quantitative Traits



What is a (mathematical) model?

-
-
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A Simple Genetic Model For Quantitative Traits



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 - $X = \bar{X} + g + e$
 - $x = g + e = 59cm$
 - how to estimate g and e ?

A Simple Genetic Model For Quantitative Traits

TABLE 19-2 Simulated Data for Days to Pollen Shed for 10 Inbred Lines of Maize Grown in Two Experiments

Experiment I

Inbred lines	A	B	C	D	E	F	G	H	I	J	Mean
Environment 1	62	64	66	66	68	68	70	70	72	74	68
Environment 2	64	66	68	68	70	70	72	72	74	76	70
Environment 3	66	68	70	70	72	72	74	74	76	78	72
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Experiment II

Inbred lines	A	B	C	D	E	F	G	H	I	J	Mean
Environment 4	58	60	62	62	64	64	66	66	68	70	64
Environment 5	64	66	68	68	70	70	72	72	74	76	70
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- $X = \bar{X} + g + e$,
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- How to calculate V_X , V_g and V_e ?

Genetic And Environment Variances

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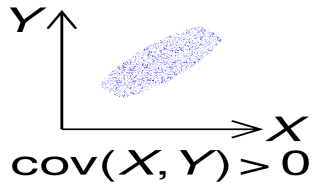
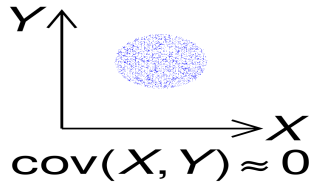
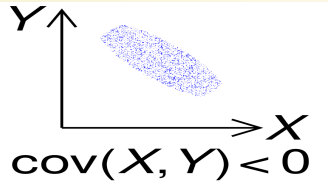
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- Introduce **Covariance**

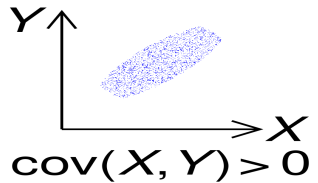
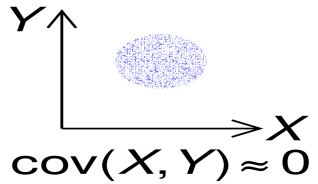
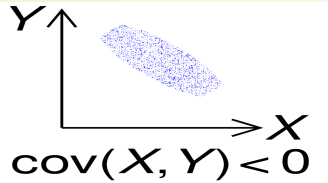
COVARIANCE

- $\text{cov}(X, Y) = E[(X - E[X])(Y - E[Y])]$



1. By Cmglee - Own work, CC BY-SA 4.0, [link](#)

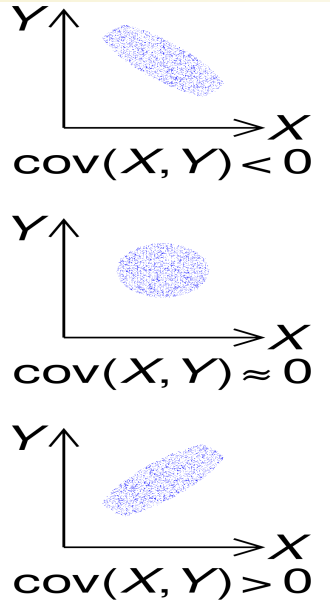
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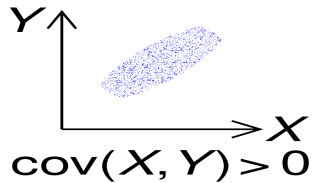
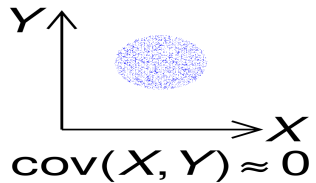
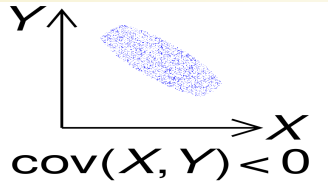
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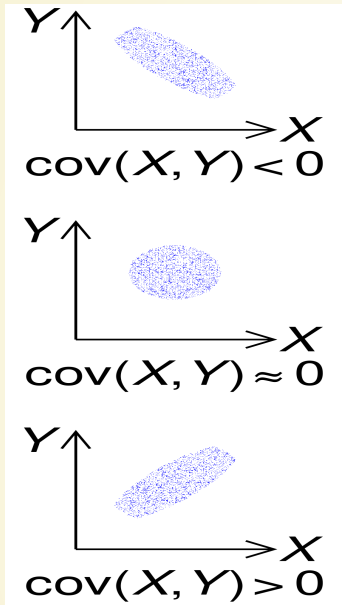
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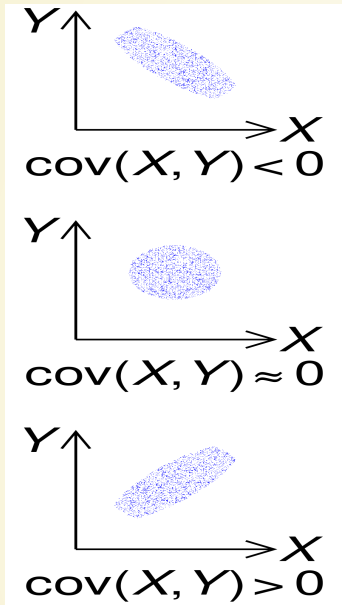
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COVARIANCE



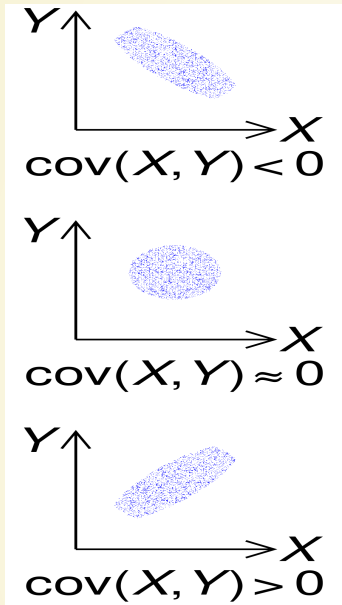
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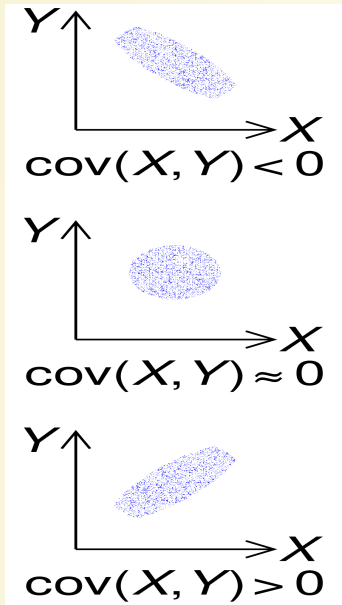
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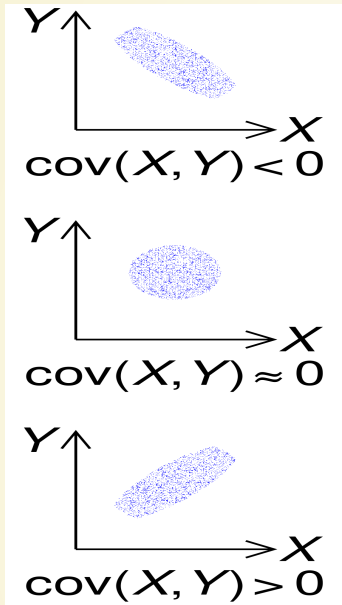
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COVARIANCE



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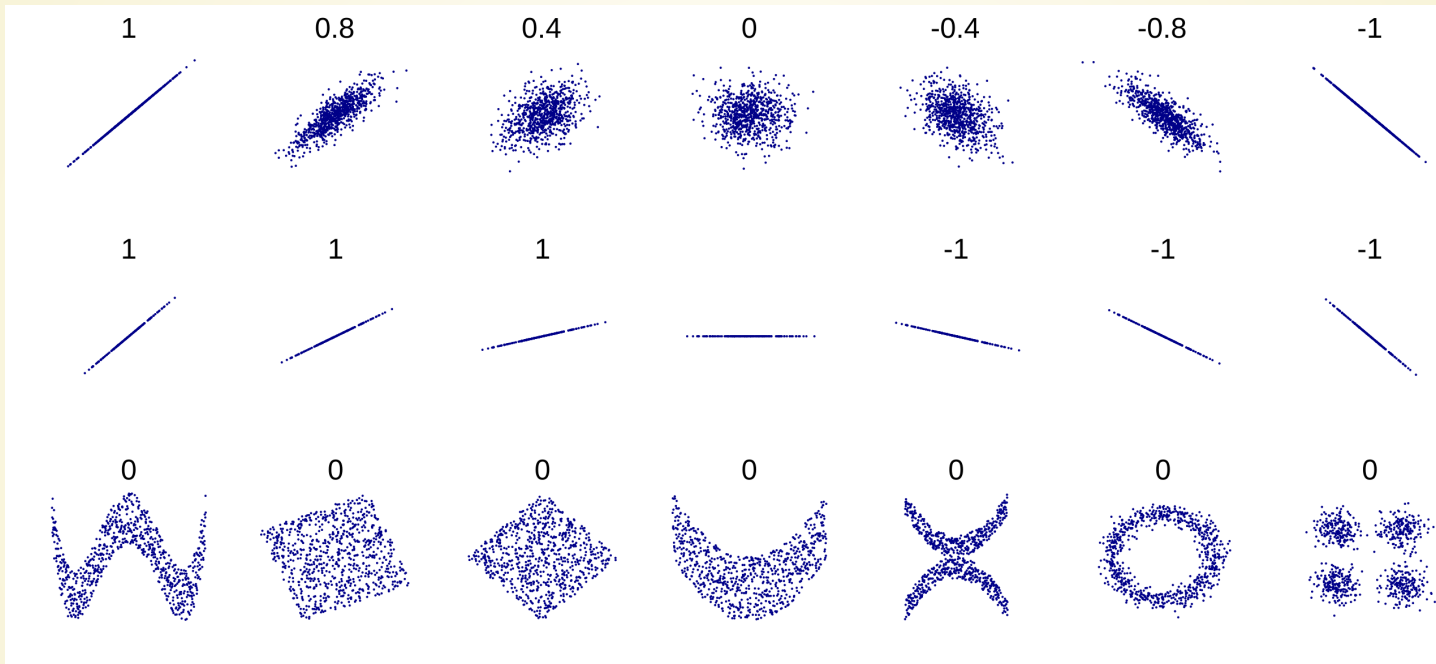
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- if g and e are correlated, we cannot use $V_X = V_g + V_e$

What Can And Can't Pearson's Correlation Coefficient Tell You?



1. By DenisBoigelot, original uploader was Imagecreator - Owggggn work, original uploader was Imagecreator, CC0, [link](#)

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- In humans, one can use **twin studies** to estimate broad sense heritability.

TABLE 19-2 Simulated Data for Days to Pollen Shed for 10 Inbred Lines of Maize Grown in Two Experiments

Experiment I

Inbred lines	A	B	C	D	E	F	G	H	I	J	Mean
Environment 1	62	64	66	66	68	68	70	70	72	74	68
Environment 2	64	66	68	68	70	70	72	72	74	76	70
Environment 3	66	68	70	70	72	72	74	74	76	78	72
Mean	64	66	68	68	70	70	72	72	74	76	70

Experiment II

Inbred lines	A	B	C	D	E	F	G	H	I	J	Mean
Environment 4	58	60	62	62	64	64	66	66	68	70	64
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- Same inbred lines, different H^2 , why?

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	Twin	
	X'	X''
1	100	110
2	125	118
3	97	90
4	92	104
5	86	89

$$H^2 = \frac{119.2}{154.3} = 0.77$$

TABLE 19-4 Broad-Sense Heritability for Some Traits in Humans as Determined by Twin Studies

Trait	H^2
Physical attributes	
Height	0.88
Chest circumference	0.61
Waist circumference	0.25
Fingerprint ridge count	0.97
Systolic blood pressure	0.64
Heart rate	0.49
Mental attributes	
IQ	0.69
Speed of spatial processing	0.36
Speed of information acquisition	0.20
Speed of information processing	0.56
Personality attributes	
Extraversion	0.54
Conscientiousness	0.49
Neuroticism	0.48
Positive emotionality	0.50
Antisocial behavior in adults	0.41
Psychiatric disorders	
Autism	0.90
Schizophrenia	0.80
Major depression	0.37
Anxiety disorder	0.30
Alcoholism	0.50–0.60
Beliefs and political attitudes	
Religiosity among adults	0.30–0.45
Conservatism among adults	0.45–0.65
Views on school prayer	0.41
Views on pacifism	0.38

Sources: J. R. Alford et al., *American Political Science Review* 99, 2005, 1–15; T. Bouchard et al., *Science* 250, 1990, 223–228; T. Bouchard, *Curr. Dir. Psych. Sci.* 13, 2004, 148–151; P. J. Clark, *Am. J. Hum. Genet.* 7, 1956, 49–54; C. M. Freitag, *Mol. Psychiatry* 12, 2007, 2–22.

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- **Additive Genetic Variance**

- the "additive effect" of an allele is defined as its **average effect** when it is paired with random alleles at its own locus and expressed across all possible genetic backgrounds (all possible alleles at other loci) in the population

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$$h^2 = \frac{V_a}{V_X} = \frac{2COV_{P,O}}{V_X}$$

